

Forest and Wildlife Resources

Social Science — Geography • Chapter 2 • Chapter Summary & Study Guide for Daksh

Did you know India has only 2.4% of world's land but supports 8% of all known species — including ~47000 kinds of plants and ~89000 kinds of animals? Such richness is called **biodiversity**, and it forms the foundation of all human life. Yet many of these species are now endangered. This chapter looks at India's forests and wildlife, why they are under threat, who is conserving them, and what role local communities play.

What is biodiversity?

Biodiversity is the wide variety of life forms in an area — plants, animals, microorganisms — and the ecosystems they form. It supports human life by giving us food, fodder, fibre, fuel, medicine, water regulation, climate moderation and the pure joy of nature. Biodiversity is interdependent — losing one species often weakens many others, like removing a brick from a wall.

IUCN classification of species

- **Normal species** — population is stable for survival. e.g. cattle, sal, pine, rodents.
- **Endangered** — survival difficult if conditions don't improve. e.g. black buck, crocodile, Indian wild ass, Indian rhino, lion-tailed macaque, sangai (Manipur deer).
- **Vulnerable** — population declining; may move to endangered if pressure continues. e.g. blue sheep, Asiatic elephant, Gangetic dolphin.
- **Rare** — small population; may move to vulnerable if conditions don't improve. e.g. Himalayan brown bear, wild Asiatic buffalo, desert fox.
- **Endemic** — found only in some areas. e.g. Andaman teal, Nicobar pigeon, Andaman wild pig.
- **Extinct** — no longer found anywhere. e.g. Asiatic cheetah, pink-headed duck.

Causes of depletion of forests and wildlife

India's forests have been shrinking dramatically since colonial times. Causes include:

- **Agricultural expansion** — between 1951 and 1980, over 26000 km² of forest was cleared.
- **Mining** — Buxa Tiger Reserve in West Bengal, dolomite mining damaged its ecosystem.
- **Large dams & projects** — submerge forests, displace wildlife.
- **Forest fires**, fuel-wood collection, overgrazing, hunting/poaching.
- **Pollution** from pesticides, untreated effluents, plastic.

How India is conserving its forests and wildlife

The Indian Wildlife (Protection) Act, 1972 set up:

- An all-India list of protected species and ban on hunting them.
- Restrictions on trade in wildlife products.
- Network of **National Parks** and **Wildlife Sanctuaries**.

Special projects have rescued specific species: **Project Tiger** (1973) — over 50 reserves today (e.g. Corbett, Sariska, Ranthambore, Sundarbans, Bandhavgarh, Periyar); **Project Elephant**, projects for rhino, lion, crocodile and great Indian bustard.

Types and distribution of forests in India

- **Reserved forests** — ~half of total forest area; most valuable for timber and conservation.
- **Protected forests** — ~one-third; protected from depletion.
- **Unclassed forests** — others (incl. those of private/community ownership in NE India and Gujarat).

Community and conservation — local heroes

Local people are often the best conservators of biodiversity:

- The **Bishnoi community** of Rajasthan reveres blackbucks and chinkaras; the famous Khejri tree martyrdom of 1730 (Amrita Devi and 363 others) saved hundreds of trees.
- **Sariska Tiger Reserve** villagers fought illegal mining.
- **Chipko Movement** in Uttarakhand (1973) — villagers, especially women, hugged trees to stop felling.
- **Sacred Groves** — patches of forest around temples or protected by faith — preserve native species (e.g. Mawphlang in Meghalaya, sacred groves of the Mundas, Bishnois).
- **Joint Forest Management (JFM)** — government partners with village communities to manage degraded forests, with locals getting a share of timber/non-timber produce.

★ MUST-REMEMBER POINTS

- India has 2.4% of world's land, 8% of species — high biodiversity.
- IUCN categories: Normal, Endangered, Vulnerable, Rare, Endemic, Extinct.
- Wildlife (Protection) Act 1972 — protected species, sanctuaries, national parks.
- Project Tiger (1973): over 50 reserves; key examples Corbett, Ranthambore, Sundarbans.
- Forests: Reserved (~half), Protected (~1/3), Unclassed.
- Community efforts: Bishnois (Khejri sacrifice), Chipko movement, sacred groves, JFM.
- Causes of biodiversity loss: agriculture, mining, dams, hunting, pollution.

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